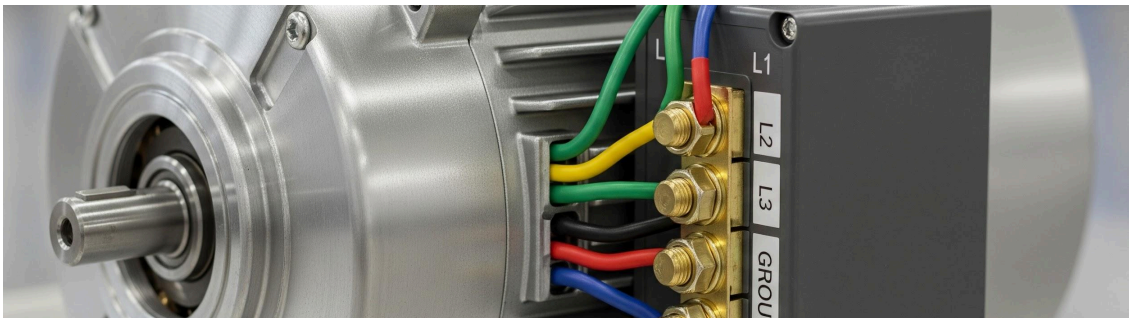

Electrical Machine Data Sheet

Machine Name/Type : DC Motor Coupled Alternator

Name plate Details

The rating section details the machine's operational capacity and design limits.

PARAMETER	DC SHUNT MOTOR	ALTERNATOR
Power output	5 HP	3.5 KVA
Voltage	220 V	415 V
Current	19 A	5 A
Speed	1500 RPM	1500 RPM
Field Current	0.8 A	1.4 A
Phase	-	3
Frequency	-	50 Hz
Extn. Volt	-	220 V
Power Factor	-	0.8

Precautions

I. Ensure **220 V DC supply** is given to the DC shunt motor before starting.

II. Never run the DC motor **without shunt field excitation**.

III. Always start the DC motor using a **starter** to limit starting current.

IV. Check proper mechanical coupling and shaft alignment.

V. Do not connect electrical load to the alternator before rated speed is achieved.

VI. Maintain correct excitation current for the alternator.

VII. Ensure proper earthing of both motor and alternator.

VIII. Do not exceed rated current of either machine.

IX. Monitor voltage, current, speed, temperature, and noise during operation.

X. Switch OFF immediately if abnormal vibration, sparking, or overheating occurs.

Working Principle of DC Shunt Motor Coupled with Alternator

The DC shunt motor works on the principle that a **current-carrying conductor placed in a magnetic field experiences a force**, producing torque and rotation.

The motor shaft is mechanically coupled to the alternator shaft. As the motor rotates at nearly constant speed, it drives the alternator.

The alternator operates on the principle of **electromagnetic induction**, where a rotating magnetic field induces an alternating voltage in the stator windings.
Thus, electrical energy is converted from **DC to mechanical and then to AC form**.

Working Characteristics

I. DC shunt motor maintains nearly constant speed under load.

II. Alternator output voltage depends on field excitation.

III. Frequency of generated AC depends on the speed of rotation.

IV. Smooth and stable operation under normal conditions.

V. Overall efficiency depends on losses in both machines.

Applications

1) Alternator characteristic studies
2) Laboratory demonstration of AC generation
3) Speed–frequency relationship experiments
4) Electrical machines training laboratories
5) Load testing of alternators
6) Educational and research setups
7) Power system laboratory experiments
8) Generator excitation studies
9) Calibration of electrical machines
10) Demonstration of electromechanical energy conversion